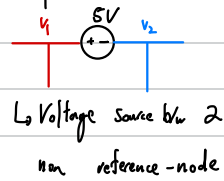
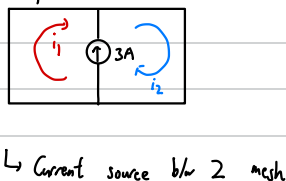


Good Luck!

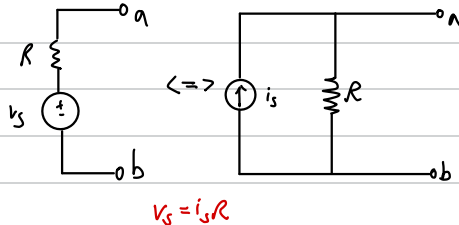
Supernode:



Supermesh:



Source transformation:



V_{TH}, I_{N}, R_{TH} :

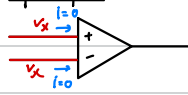
1. Turn off all independent sources
voltage $\rightarrow$ short, current $\rightarrow$ open
2. Find R_{eq} if only indep.
3. Apply test source

Maximum Power transfer:

$$P_{max} = V_{TH}^2 / 4R_{TH} = I_{N}^2 R_{TH} / 4$$

$$R_L = R_{TH}$$

Op Amps:



Resistance (R):

$$R = \frac{\rho L}{A}$$

ρ - Resistivity
 L - length
 A - Surface Area

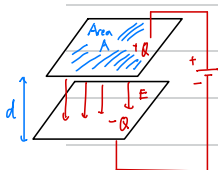
Time constant, τ :

Time required for response to decay to k (36.8%)

RMS formula:

$$X_{rms} = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt}$$

Parallel Plate Capacitor



voltage cannot "jump"
 $\hookrightarrow V_c = C \frac{dv}{dt} \neq \infty$

$\hookrightarrow$ Current:

$$i(t) = C \frac{dv}{dt}$$

$\hookrightarrow$ Voltage:

$$v(t) = \frac{1}{C} \int i(t) dt + v(0)$$

$\hookrightarrow$ Power:

$$p(t) = i(t)v(t)$$

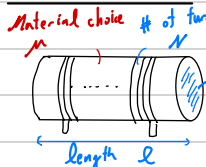
$\hookrightarrow$ Energy

$$w = \frac{1}{2} C v^2 = \frac{Q^2}{2C}$$

$$E, \text{ electric field} = \frac{V_0}{d} = \frac{Q}{A \epsilon_r \epsilon_0}$$

$$C = \frac{Q}{V_0} = \frac{A \epsilon_r \epsilon_0}{d} \left[F = \frac{\text{Coulombs}}{\text{Volts}} \right]$$

Solenoid Inductors



$$\text{Magnetic flux density} = \frac{\mu_r \mu_0 N^2}{l} i \text{ [Tesla]}$$

$$\Phi_m, \text{ Magnetic flux} = B(A) \text{ [Wb]}$$

$$L = N \frac{\Phi_m}{i} = \frac{\mu_r \mu_0 N^2 A}{l} \left[H = \frac{Vs}{A} \right]$$

Cross-sectional Area, A

current cannot "jump"
 $\hookrightarrow v = L \frac{di}{dt} \neq \infty$

$\hookrightarrow$ Current:

$$v(t) = L \frac{di}{dt}$$

$\hookrightarrow$ Voltage:

$$i(t) = \frac{1}{L} \int v(t) dt + i(0)$$

$\hookrightarrow$ Power:

$$p(t) = v(t)i(t)$$

$\hookrightarrow$ Energy:

$$w(t) = \frac{1}{2} L i^2$$

Source free RC:

$$v(t) = V_0 e^{-t/\tau}$$

$$\tau = RC$$

Source free RL:

$$i(t) = I_0 e^{-t/\tau}$$

$$\tau = \frac{L}{R}$$

Solving First-order circuit:

Step 1: Draw circuit for $t < 0s$, find $i_L(0^-) = i_L(0^+)$ or $v_C(0^+) = v_C(0^-)$

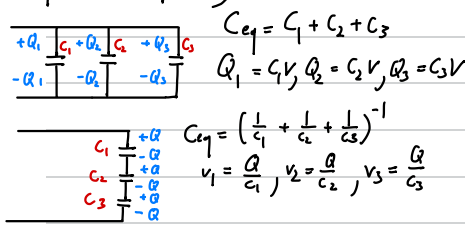
Step 2: Draw circuit for $t = (0^+)s$, replace inductor with current source $i = i_L(0^+)$, find $v(0^+)$ or $i(0^+)$
replace capacitor with voltage source $v = v_C(0^+)$

Step 3: Draw circuit $t > 0s$, determine R_{TH} of inductor/capacitor

Find τ from this, $\tau = \frac{L}{R_{TH}}$ for RL, $\tau = R_{TH}C$ for RC

Step 4: Draw circuit $t \rightarrow \infty$, replace inductor with short circuit, replace capacitor with open circuit, find $i(\infty)$ or $v(\infty)$
And $v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$, $i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$

Equivalent Capacitance, Inductance



Sinusoids, Phasors

$\hookrightarrow$ Conversions

$$-j = \frac{1}{j}, j = 1 \angle 90^\circ, j = 1 \angle -90^\circ$$

$z(j\omega) \rightarrow$ Differentiation

$z(j\omega) \rightarrow$ Integration

$$v(t) = V_0 \sin(\omega t + \theta) \rightarrow \bar{V} = V_0 \angle \theta - \frac{\pi}{2}$$

$$i(t) = I_0 \cos(\omega t + \theta) \rightarrow \bar{I} = I_0 \angle \theta$$

$$\sin(x + \theta) = \cos(x + \theta - 90^\circ), \cos(x + \theta) = \sin(x + \theta + 90^\circ)$$

$$\sin(x + \theta) = -\sin(x + \theta - 180^\circ), \cos(x + \theta) = -\cos(x + \theta - 180^\circ)$$

$\hookrightarrow$ Phasor operations

$$\text{for } z = x + jy = r \angle \theta = r e^{j\theta}$$

$$z_1 z_2 = r_1 r_2 \angle \theta_1 + \theta_2$$

$$z_1 / z_2 = r_1 / r_2 \angle \theta_1 - \theta_2$$

$$\sqrt{z} = \sqrt{r} \angle \theta / 2$$

$$z^* = x - jy = r \angle -\theta$$

$$z \angle -\theta = -z \angle \theta + 180^\circ$$

$\hookrightarrow$ Leading, lagging

$$\text{for } v_A = V_0 \sin(\omega t + \theta_A) = V_0 \angle \theta_A$$

$$v_B = V_0 \sin(\omega t + \theta_B) = V_0 \angle \theta_B$$

If $\theta_A > \theta_B$, v_A leads v_B

$\theta_A < \theta_B$, v_A lags v_B

$\hookrightarrow$ frequency, $f = \frac{1}{T}$

angular frequency $\omega = 2\pi f$

Impedance, Admittance

In General, $G(\omega) \neq \frac{1}{R(\omega)}, B(\omega) \neq \frac{1}{X(\omega)}$

$$\text{Phasor } \bar{Z} \text{ (Impedance)} = \bar{V} / \bar{I}$$

$$\bar{Z} = R(\omega) + jX(\omega) \text{ (Units } \Omega \text{)}$$

$$R(\omega): \text{ Resistance, } = \text{Re}\{\bar{Z}\}$$

$$X(\omega): \text{ Reactance, } = \text{Im}\{\bar{Z}\}$$

$$\text{Phasor } \bar{Y} \text{ (Admittance)} = \bar{I} / \bar{V}$$

$$\bar{Y} = G(\omega) + jB(\omega) \text{ (Units } S \text{)}$$

$$G(\omega): \text{ Conductance } = \text{Re}\{\bar{Y}\}$$

$$B(\omega): \text{ Susceptance } = \text{Im}\{\bar{Y}\}$$

$\hookrightarrow$ Impedance of Resistor, Capacitor, Inductor:

$$\bar{Z}_R = V / I \angle 0^\circ$$

$$\bar{Z}_C = V / I \angle -90^\circ = -j / \omega C = \frac{1}{\omega C} \angle -90^\circ$$

$$\bar{Z}_L = V / I \angle 90^\circ = j\omega L = \omega L \angle 90^\circ$$

Power equations:

↳ Instantaneous Power, *function of time*

$$p(t) = v(t)i(t) = \frac{1}{2} V_0 I_0 \cos(\theta_v - \theta_i) + \frac{1}{2} V_0 I_0 \cos(2\omega t + \theta_v + \theta_i)$$

avg *oscillating*

↳ Angle, root mean squared

$$\theta_z = \theta_v - \theta_i = \theta_{rms} - \theta_{irms}$$

$$\bar{V}_{rms} = \bar{V}/\sqrt{2}, V_{rms} = V_0/\sqrt{2}$$

$$\bar{I}_{rms} = \bar{I}/\sqrt{2}, I_{rms} = I_0/\sqrt{2}$$

↳ Maximum *(real)* average power

When $\bar{Z}_L = \bar{Z}_{TH}^* = R_{TH} - jX_{TH}$

$$P_{max} = |\bar{V}_{TH}|^2 / 8 R_{TH} \text{ (Watts, W)}$$

↳ If load, R is purely resistive,
 $R = |\bar{Z}_{TH}|$

↳ Complex power, *phasor*

$$\bar{S} = \frac{1}{2} \bar{V} \bar{I}^*$$

$$= \bar{V}_{rms} \bar{I}_{rms}^*$$

$$= |\bar{V}_{rms}|^2 / \bar{Z}_L^*$$

$$= |\bar{I}_{rms}|^2 \bar{Z}_L$$

$$= V_{rms} I_{rms} \angle \theta_z = |\bar{V}_{rms}| |\bar{I}_{rms}| \angle \theta_z$$

$$= V_{rms} I_{rms} \cos(\theta_z) + j V_{rms} I_{rms} \sin(\theta_z)$$

$$= S \cos(\theta_z) + j S \sin(\theta_z)$$

$$= P + j Q \text{ (Volts-Ampere, VA)}$$

↳ Apparent power, *scalar*

$$S = |\bar{S}|$$

$$= \frac{1}{2} V_0 I_0 \text{ (} V_0, I_0 \text{ are amplitudes or peaks)}$$

$$= V_{rms} I_{rms} = |\bar{V}_{rms}| |\bar{I}_{rms}|$$

$$= \sqrt{P^2 + Q^2} \text{ (Volts-Ampere, VA)}$$

↳ Real (or average) power, *scalar*

$$P = \frac{1}{2} V_0 I_0 \cos(\theta_z)$$

$$= V_{rms} I_{rms} \cos(\theta_z) = |\bar{V}_{rms}| |\bar{I}_{rms}| \cos \theta_z$$

$$= S \cos(\theta_z) \quad \cos \theta_z = pf$$

$$= \text{Re}\{\bar{S}\} \text{ (Watts, W)}$$

↳ Power factor, *scalar*

$$pf = \cos \theta_z$$

$$= P/S \text{ (No units)}$$

↳ Reactive power, *scalar*

$$Q = \frac{1}{2} V_0 I_0 \sin(\theta_z)$$

$$= V_{rms} I_{rms} \sin(\theta_z) = |\bar{V}_{rms}| |\bar{I}_{rms}| \sin \theta_z$$

$$= S \sin(\theta_z)$$

$$= \text{Im}\{\bar{S}\} \text{ (Volts-Ampere-Reactive, VAR)}$$

Purely resistive loads:

$$Q = 0$$

$$P = V_{rms} I_{rms}$$

$$\bar{S} = P$$

$$pf = 1 \text{ "unity pf"}$$

Purely capacitive loads:

$$P = 0$$

$$Q = -V_{rms} I_{rms}$$

$$\bar{S} = -j V_{rms} I_{rms}, \theta_z = -90^\circ$$

$$pf = 0 \text{ "leading"}$$

voltage lags current by 90°

Purely inductive loads:

$$P = 0$$

$$Q = V_{rms} I_{rms}$$

$$\bar{S} = j V_{rms} I_{rms}, \theta_z = 90^\circ$$

$$pf = 0 \text{ "lagging"}$$

voltage lags current by 90°

Inductive loads:

$$Q > 0$$

$$pf \rightarrow \text{lagging} > 0$$

current lags voltage, $\theta_z > 0$

Capacitive loads:

$$Q < 0$$

$$pf \rightarrow \text{leading} > 0$$

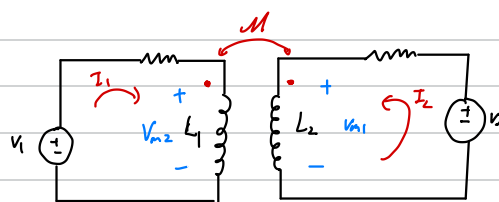
current leads voltage, $\theta_z < 0$

pf correction:

To change $pf_1 = \cos \theta_1 \rightarrow pf_2 = \cos \theta_2$
with given Avg Power P ,

$$C = \frac{P(\tan \theta_1 - \tan \theta_2)}{\omega V_{rms}^2}$$

Mutual inductance:

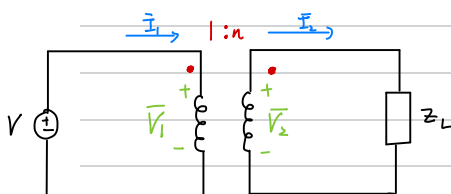


$$\bar{V}_2 = j\omega M \bar{I}_1, \bar{V}_1 = j\omega M \bar{I}_2$$

$$\bar{Z}_M = j\omega M$$

Mutual inductance, M
 $M_{21} = M_{12} = k \sqrt{L_1 L_2} \rightarrow k$
 k is coupling coeff.

Ideal transformers:



$$\text{Turns ratio, } n = \frac{N_2}{N_1}$$

$$\frac{V_2}{V_1} = \pm n = \pm \frac{N_2}{N_1}$$

$$\frac{I_2}{I_1} = \mp \frac{1}{n} = \mp \frac{N_1}{N_2}$$

↳ Dot convention:

Current 1, I_1 enters through dot $\rightarrow$ Voltage 2, V_2 +ve at dot
Current 1, I_1 leaves from dot $\rightarrow$ Voltage 2, V_2 -ve at dot

↳ Dot convention:

If V_1, V_2 both +ve or both -ve at dot $\rightarrow \bar{V}_2 / \bar{V}_1 = +n = +N_2 / N_1$
else $\rightarrow \bar{V}_2 / \bar{V}_1 = -n = -N_2 / N_1$

If I_1, I_2 both enter or leave the dot $\rightarrow \bar{I}_2 / \bar{I}_1 = -1/n = -N_1 / N_2$
else $\rightarrow \bar{I}_2 / \bar{I}_1 = +1/n = +N_1 / N_2$

↳ Conservation of complex power

$$\bar{S}_1 = \bar{V}_1 \bar{I}_1^* = \bar{V}_2 \bar{I}_2^* = \bar{S}_2$$

↳ Impedance "matching"

$$Z_{in} = Z_L / n^2 = \bar{V}_1 / \bar{I}_1 = k^2 \frac{V_2}{I_2}$$